The DADA Options Pricer

Tail-Risk Hedging in Non-Stationary Regimes

Harry J. Potter

Department of Mathematics, Hogwarts School of Witchcraft and Wizardry

harry.potter@example.com · April 2026

Abstract. We present DADA, a unified options-pricing toolkit combining the Black-Scholes-Merton closed-form solver with a calibrated Heston stochastic-volatility model and a 1,000,000-path Monte Carlo engine. The framework is motivated by the observation that catastrophic-tail events ("dementor moves") are systematically mispriced by Gaussian models when the volatility regime is non-stationary. We benchmark the pricer against 14 years of S&P 500 option chains and find that the Heston-calibrated implementation reduces hedging error by 31% in the left tail (returns below the 1st percentile) versus a Black-Scholes baseline, at a marginal computational cost of 4.2× wall-clock time.

1. INTRODUCTION

The pricing of options under non-stationary volatility regimes remains one of the most practically important open problems in quantitative finance. While the Black-Scholes-Merton framework [1] provides a tractable analytical baseline, its assumption of constant volatility is empirically rejected by virtually every traded options surface. The implied-volatility smile, the leverage effect, and volatility clustering are all stylised facts that demand a richer model.

This paper documents the *Defence Against the Dark Arts* (DADA) pricer, an independent-study capstone built over the 2024-2025 academic year. DADA is designed for three use cases: (i) rapid mark-to-model valuation across a vanilla options book, (ii) Greek computation for delta- and vega-hedging, and (iii) Monte Carlo path generation for path-dependent payoffs and tail-risk scenario analysis.

2. MODELS

2.1 Black-Scholes-Merton baseline

Under the standard risk-neutral measure, the price of a European call satisfies

$$C(S, t) = S \cdot N(d_1) - K \cdot e^{-r(T-t)} \cdot N(d_2)$$

where $N(\cdot)$ denotes the standard normal CDF and d_1, d_2 are the canonical Black-Scholes arguments. We use this solver as a benchmark and as a sanity check for the Monte Carlo engine.

2.2 Heston stochastic volatility

We adopt the standard Heston [2] dynamics for the underlying S_t and its instantaneous variance v_t:

$$dS_t = r\, S_t\, dt + \sqrt{v_t}\, S_t\, dW_t^S$$

$$dv_t = \kappa(\theta - v_t)\, dt + \sigma \sqrt{v_t}\, dW_t^v$$

with correlation ρ between the two Brownian motions. Calibration is performed by minimising squared implied-vol errors across 32 strikes and 6 maturities, using the SciPy `least_squares` trust-region solver with analytic Jacobians where available.

2.3 Monte Carlo engine

Paths are simulated using a full-truncation Euler scheme on the log-price with antithetic variates and Sobol' quasi-random sequences for variance reduction. The default configuration generates 1,000,000 paths over 252 daily steps in approximately 11 seconds on a single M3-class CPU core; GPU acceleration via CuPy is available as an optional path.

3. EMPIRICAL RESULTS

We benchmark all three implementations against 14 years of daily S&P 500 option chains (2010-2024). Hedging error is measured as the realised mean-squared error of a daily delta-rebalanced hedge over the option's lifetime. Table 1 summarises the headline findings.

Model	Mean MSE	P99 MSE	Left-tail error	Wall-clock
Black-Scholes	1.00 (baseline)	1.00	1.00	1.0×
Heston (calibrated)	0.78	0.74	**0.69**	4.2×
Monte Carlo, 1M paths	0.82	0.79	0.71	9.6×

Table 1: Hedging error normalised to Black-Scholes baseline (lower is better). Left-tail error is the MSE conditional on the underlying realising a return below the 1st percentile of its rolling 252-day distribution.

The 31% reduction in left-tail hedging error is the headline result of the project. Crucially, the improvement is concentrated precisely in the regime where Black-Scholes hedges fail catastrophically — that is, exactly when a hedger most needs the model to work.

4. LIMITATIONS AND FUTURE WORK

Three limitations are worth flagging. First, the Heston calibration is performed on end-of-day mid-quotes; intraday recalibration would likely improve the fit but introduces overfitting risk. Second, the model assumes a constant risk-free rate; extending to a stochastic short-rate (Hull-White) is

straightforward but was deferred. Third, jumps are absent — a Bates [3] or SVCJ extension is the natural next step and is currently in development as a follow-on independent study.

5. CONCLUSION

DADA demonstrates that a carefully calibrated stochastic-volatility model materially outperforms the Black-Scholes baseline in precisely the tail regime that matters for risk management. The full implementation is open-sourced at `github.com/clg236/dada-pricer` under an MIT licence.

REFERENCES

[1] Black, F. and Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3), 637-654.

[2] Heston, S. L. (1993). A Closed-Form Solution for Options with Stochastic Volatility. *Review of Financial Studies*, 6(2), 327-343.

[3] Bates, D. S. (1996). Jumps and Stochastic Volatility. *Review of Financial Studies*, 9(1), 69-107.

[4] Gatheral, J. (2006). *The Volatility Surface: A Practitioner's Guide*. Wiley.

[5] Glasserman, P. (2004). *Monte Carlo Methods in Financial Engineering*. Springer.